TMM4250 Project work, Scissor lift simulation in ABAQUS, v1

October 2026



Figure 1: Example of scissor-lift table

Design and analyze a scissor lift table as illustrated in the Figure 1. The primary use is lifting a rather "beefy" Harley Davidson motorcycle.

Overall requirements

Lift range 200-1000mm, i.e.

a. Maximum height at lowest position 200mm

b. Minimum height at highest position is 1000mm

Basic assessments

For you selected actuator (hydraulic cylinder or similar) establish the following;

1. Lift capacity as a function of lift height, i.e., as a function of cylinder force or pressure.
2. Buckling load as a function of lift height.
3. Estimate sensitivity with respect to eccentric loading. Loading inside support surface should not give failure.
4. Eigen-frequencies as a function of lift height and lift load (mass).

Idle speed of a Harley Davidson is in the range 800-1200 rpm. Eigen-frequencies in this range should be avoided.

Local design solutions *(Optional)*

Table top or working surface should be designed with concentrated forces in mind. What is necessary load area in order to avoid

1. Permanent deformation

2. Penetration/failure.

Suggested designs to evaluate:

a. Plywood (cross-ply laminated wood) with or without thin metal surfacing.

b. Solid or stiffened metal plate in steel or aluminum.

c. GRP/sandwich panel.

Detailed design of joints*(Optional)*

Perform necessary 3D detailed analysis of the joints and compare with the simplified connectors from the global analyses.

Reporting

The work should be reported as a technical report. Document your choices with respect to element types, analysis types, etc. The report should have a sufficient level of detail so that other engineers would be able to reproduce and verify your results.